TRINETRA

AI-Powered Digital Public Safety Intelligence Platform

ET AI Hackathon 2.0 | Problem Statement 6

AI for Digital Public Safety: Defeating Counterfeiting, Fraud & Digital Arrest Scams

Submission Detail	Information
Participant	RITHISH PK , CHINMAY DEEPAK CHANDAVAR
Problem Statement	PS 6 AI for Digital Public Safety
Live Demo	https://trinetra-india.netlify.app
GitHub Repository	https://github.com/RithishPK/trinetra
Backend API	https://trinetra-backend-209a.onrender.com
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1. Executive Summary

Trinetra is a real-time AI-powered Digital Public Safety Intelligence Platform designed to protect Indian citizens from digital fraud before financial loss occurs. Built for ET AI Hackathon 2.0 under Problem Statement 6, Trinetra addresses three critical threat vectors digital arrest scams, message-based fraud, and fake document verification through a citizen-facing web platform powered by Llama 3.3 70B via Groq API.

India registered 1.14 million cybercrime complaints in 2023, a 60% increase from 2022. The Ministry of Home Affairs reported that digital arrest scams alone defrauded citizens of over Rs 1,776 crore in just the first nine months of 2024. Trinetra shifts the response from reactive complaint filing to proactive, real-time fraud detection accessible to every Indian citizen regardless of technical literacy.

Key achievements of this submission:

- Three live AI detection modules with calibrated precision and recall
- Deterministic verdict logic verdicts computed in code, not by LLM, eliminating hallucination
- Multilingual detection Hindi, Tamil, Telugu and other Indian languages supported
- Court-ready audit trail unique Case ID, UTC timestamp, and platform signature on every analysis
- Community intelligence layer live scam reporting feed powered by Supabase PostgreSQL
- Mobile-responsive design works on any device without installation
- Fully deployed live on Netlify (frontend) and Render (backend)

2. Problem Statement Analysis

2.1 Scale of the Crisis

India faces an industrialised digital fraud epidemic. According to the Indian Cyber Crime Coordination Centre (I4C), more than 92,000 digital arrest scam cases were reported in India in the first ten months of 2024, with victims losing an estimated Rs 2,140 crore. The number of reported cases nearly tripled between 2022 and 2024.

These are not opportunistic crimes. They are structured operations run from fraud compounds in Southeast Asia Thailand, Cambodia, and Myanmar using spoofed numbers, AI-generated voices, fake government portals, and deepfake video technology. Indian nationals are frequently trafficked into these compounds and forced to operate fraud scripts.

2.2 Key Threat Vectors Addressed

Digital Arrest Scams

Fraudsters impersonating CBI, ED, Customs, TRAI, or Police officers contact victims via phone or video call, falsely claim their Aadhaar or mobile number is linked to criminal activity, and psychologically trap them in "digital arrests" demanding money transfers to avoid arrest. The attack chain is consistent: impersonation, psychological pressure, isolation from family, financial extraction, and technical spoofing via fake police station backgrounds and forged documents.

Message and Communication Fraud

WhatsApp, SMS, and Telegram channels are weaponised for fake job offers with registration fees, fake KYC expiry threats from banks, fraudulent investment schemes promising unrealistic returns, task scams offering money for video likes, and impersonation of government agencies and known companies.

Fake Document Fraud

Forged government notices, fake arrest warrants, fraudulent employment offer letters, and fake customs clearance notices are circulated to establish legitimacy before financial demands are made. These documents are designed to look official enough to bypass casual inspection.

2.3 What Law Enforcement Currently Lacks

As stated in the PS: "What law enforcement lacks is not evidence after the fact it is intelligence before mass victimisation occurs, and reliable tools to detect threats at the point of contact rather than the point of complaint."

Trinetra directly addresses this gap by providing citizens with a tool that flags fraud at the point of first contact before any money is transferred or personal data is shared.

3. Solution: Trinetra Platform

3.1 Platform Overview

Trinetra is a citizen-facing web platform that provides real-time AI fraud detection across three threat vectors, plus a community intelligence layer. The platform is accessible via any browser without installation, works on mobile devices, and requires no technical expertise from the user.

Live URL: https://trinetra-india.netlify.app

3.2 Detection Modules

Module 1: Digital Arrest Scam Detector

Input: Text (call script, WhatsApp message, SMS) in any Indian language or English.

The detector evaluates five signal dimensions:

- Impersonation signals claims of CBI, ED, Customs, Police, TRAI, RBI, or Supreme Court authority
- Psychological pressure signals threats of arrest, FIR, money laundering charges, passport cancellation
- Isolation tactics instructions to not tell family, stay on call, keep matter confidential
- Financial extraction signals security deposit, clearance fee, UPI transfer, bank account requests
- Technical spoofing signals fake video call setup, badge display, staged police station backgrounds

Scoring bands: 0-20 SAFE | 21-45 LOW RISK | 46-70 MODERATE RISK | 71-89 HIGH RISK | 90-100 CONFIRMED SCAM

Multilingual: If input is in Hindi, citizen advice is returned in Hindi. Tamil input returns Tamil advice. This is enforced via language detection in the system prompt.

Module 2: Message Fraud Detector

Input: Text (WhatsApp, SMS, Telegram message).

Classifies across 8 fraud taxonomy categories:

- Job Scam registration fees, guaranteed income, no interview
- Investment Scam fake trading apps, crypto schemes, guaranteed returns
- KYC Scam fake bank/UPI KYC expiry, account suspension threats
- Lottery/Prize Scam fake wins requiring fee payment
- Impersonation Scam fake government, bank, or company representative
- Romance/Trust Scam relationship building leading to financial request
- Task Scam "like videos and earn money" micro-task platforms
- Unknown suspicious but does not fit above categories

Red flag detection includes: registration fees, guaranteed income claims, no company name, personal WhatsApp numbers, urgency language, Aadhaar/PAN requests before interview.

Scoring bands: 0-20 SAFE | 21-65 SUSPICIOUS | 66-85 LIKELY SCAM | 86-100 DEFINITE SCAM

Module 3: Document Verifier

Input: PDF upload (government notice, offer letter, legal document).

Analyses six document categories:

- Fake Government Notice impersonating CBI, ED, IT Department, Police, Courts, TRAI, RBI
- Fake Legal Summons fake court orders, arrest warrants, legal notices demanding money
- Fake Employment Offer forged offer letters from known companies
- Fake KYC/Banking Document fake bank notices, fake RBI circulars
- Fake Customs/Courier Notice "package detained," clearance fee demanded
- Legitimate Document document shows no significant fraud indicators

Text extraction via pdfplumber. Structural, linguistic, and content red flags evaluated. Analysis limitation disclosed: visual security features cannot be verified from text alone.

Scoring bands: 0-25 LIKELY LEGITIMATE | 26-50 SUSPICIOUS | 51-75 LIKELY FRAUDULENT | 76-100 CONFIRMED FRAUDULENT

Module 4: Community Scam Feed

Citizens report scams they have encountered via a submission form specifying scam type, state, city, and description. Reports are stored in Supabase PostgreSQL with Row Level Security. A live feed displays the 20 most recent reports from across India, creating collective intelligence from individual experiences.

This partially addresses the "fraud network detection" requirement by aggregating citizen intelligence in real time.

4. Technical Architecture

4.1 System Architecture

Trinetra follows a clean three-tier architecture with a React frontend, FastAPI backend, and Groq LLM inference layer, with Supabase as the community data store.

Layer	Technology	Deployment
Frontend	React 18, Tailwind CSS, Vite, Axios	Netlify (CDN)
Backend	FastAPI, Python 3.14, pdfplumber, uvicorn	Render (Free tier)
LLM Inference	Llama 3.3 70B via Groq API	Groq Cloud
Database	Supabase PostgreSQL	Supabase (Singapore)
Version Control	Git, GitHub	github.com/RithishPK/trinetra

4.2 API Endpoints

Endpoint	Method	Input	Description
/api/digital-arrest/analyze	POST	JSON {text}	Analyse digital arrest scam text
/api/message-fraud/analyze	POST	JSON {text}	Analyse message fraud text
/api/document-verify/analyze	POST	multipart/form-data PDF	Analyse uploaded PDF document
/api/scam-report/report	POST	JSON report object	Submit community scam report
/api/scam-report/feed	GET	None	Retrieve 20 latest community reports

4.3 Prompt Engineering Approach

All three detection modules use few-shot prompt engineering on Llama 3.3 70B. Key design decisions:

- System prompt includes real-world context from MHA advisories and documented Indian scam cases
- Few-shot examples extracted from real Indian scam transcripts, news reports, and PIB fact-checks
- Model instructed to reason through signal dimensions before scoring chain-of-thought approach
- Hallucination guard: model explicitly instructed to only flag signals directly evidenced in text
- Language detection instruction: citizen advice returned in detected input language
- Temperature set to 0.1 for consistent, deterministic outputs

4.4 Deterministic Verdict Logic

A critical architectural decision: verdicts are NOT assigned by the LLM. The LLM outputs only a numeric risk score (0-100). The verdict label is then computed deterministically in Python code based on predefined score bands. This eliminates the common failure mode where LLMs assign inconsistent verdicts that contradict the numeric score.

Example (digital arrest router):

```
if score <= 20: verdict = "SAFE"
elif score <= 45: verdict = "LOW_RISK"
elif score <= 70: verdict = "MODERATE_RISK"
elif score <= 89: verdict = "HIGH_RISK"
else: verdict = "CONFIRMED_SCAM"
```

4.5 Audit Trail Implementation

Every analysis result includes:

- case_id unique identifier in format TRI-XXXXXXXX (8-character UUID fragment)
- analyzed_at UTC ISO 8601 timestamp
- platform "Trinetra v1.0"

This audit trail addresses the PS requirement for "auditability of intelligence packages for legal admissibility." Each analysis can be referenced by its Case ID for follow-up reporting.

5. Alignment with Judging Criteria

Criterion	Weight	Trinetra Approach
Innovation	25%	Deterministic verdict logic, multilingual detection, community intelligence layer, audit trail for legal admissibility
Business Impact	25%	Addresses Rs 1,776Cr fraud problem, citizen-first design, zero friction (no login), deployable via any browser
Technical Excellence	20%	Production-deployed stack, calibrated prompts with real Indian scam data, few-shot engineering, clean API design
Scalability	15%	Stateless FastAPI backend, Supabase scales horizontally, Groq API handles concurrent inference, Phase 2/3 roadmap defined
User Experience	15%	Mobile-responsive, dark theme, instant results, multilingual advice, clear verdict with actionable guidance

5.1 Evaluation Focus Alignment

PS Evaluation Focus	Trinetra Coverage
Digital arrest scam detection precision and recall	Calibrated on real MHA/CBI impersonation patterns. Tested: CONFIRMED SCAM at 98. Legitimate MHA advisory: SAFE at 0.
False positive rate for citizen-facing tools	Deterministic verdict logic. Real MHA advisory scores 0. Real Infosys offer scores 8. Tuned to <2% false positive target.
Auditability for legal admissibility	Every result has unique Case ID, UTC timestamp, and platform signature. Red flags are evidence-only no inference without text evidence.
Fraud network detection lead time	Community scam feed aggregates citizen reports in real time, providing early warning of new scam patterns by type and geography.
Counterfeit detection accuracy	Not implemented in Phase 1 (honest limitation). Planned for Phase 3 using YOLOv8 with RBI-provided FICN training data.

6. Test Results and Demo Evidence

6.1 Digital Arrest Scam Detector

Test Input	Expected	Actual Verdict	Score	Pass/Fail
Hindi CBI impersonation + security bond + isolation	CONFIRMED SCAM	CONFIRMED SCAM	98	PASS
Full digital arrest script with UPI demand	CONFIRMED SCAM	CONFIRMED SCAM	92	PASS
Legitimate MHA advisory about digital arrest scams	SAFE	SAFE	0	PASS
Real bank fraud monitoring team call-back request	LOW RISK	LOW RISK	30	PASS

6.2 Message Fraud Detector

Test Input	Expected	Actual Verdict	Score	Pass/Fail
Fake Amazon job + Rs 500 fee + WhatsApp number	DEFINITE SCAM	DEFINITE SCAM	95	PASS
Guaranteed job without interview + Aadhaar request	LIKELY SCAM	LIKELY SCAM	75	PASS
Genuine Infosys job application acknowledgement	SAFE	SAFE	8	PASS
Vague recruiter message with personal WhatsApp	SUSPICIOUS	SUSPICIOUS	40	PASS

6.3 Document Verifier

Test Input	Expected	Actual Verdict	Score	Pass/Fail
Fake ED arrest warrant with UPI payment demand	CONFIRMED FRAUDULENT	CONFIRMED FRAUDULENT	80	PASS
Axlero offer letter (legitimate structure, scam onboarding)	LIKELY LEGITIMATE	LIKELY LEGITIMATE	20	PASS*

*Note: The Axlero offer letter scores as LIKELY LEGITIMATE because the Rs 500 onboarding fee demand occurs in a separate communication, not in the PDF itself. This is a documented limitation of text-only PDF analysis disclosed in all results.

7. Known Limitations and Honest Disclosure

7.1 Current Limitations

- Counterfeit currency detection is not implemented. Building a credible CV model for currency detection requires RBI-provided FICN datasets and significant training infrastructure beyond a hackathon timeline.
- Geospatial fraud mapping is not implemented. Community scam reports capture state and city but do not render on a map in Phase 1.
- Fraud network graph intelligence is not implemented. Transaction metadata analysis requires law enforcement data access.
- Document verifier is text-only. Visual security features, seals, watermarks, and signatures cannot be verified from extracted PDF text.
- Scams that embed payment demands in separate onboarding processes outside the document cannot be detected from PDF analysis alone.
- Render free tier spins down after inactivity first request may have 50-second cold start delay.
- Evaluation dataset of 20 messages per module larger validation sets would strengthen precision/recall claims.

7.2 Why These Limitations Are Acceptable

The PS explicitly states that the listed areas "are illustrative only." Trinetra chooses depth over breadth delivering production-quality citizen protection for the highest-impact use case (digital arrest scams) rather than shallow coverage of all five suggested areas. The evaluation criteria weights Innovation (25%) and Business Impact (25%) highest areas where Trinetra's focused, deployable approach scores strongly.

8. Scalability and Phase Roadmap

Phase 1 Current Submission

- Digital arrest scam detection (text, multilingual)
- Message fraud detection (8 categories)
- PDF document verification
- Community scam reporting feed
- Audit trail with Case ID
- Mobile-responsive web platform
- Deployed on Netlify + Render

Phase 2 Next 6 Months

- WhatsApp Business API integration citizens can forward suspicious messages directly to Trinetra bot
- IVR system toll-free number citizens can call to verify a suspicious call in real time
- 12 regional language support expand from current 3-4 to all scheduled Indian languages
- Telecom API integration flag known scam numbers before citizens answer
- Automated MHA alert generation high-confidence detections trigger alerts to cybercrime portal
- NCRB portal integration guided one-click reporting flow

Phase 3 12-18 Months

- Counterfeit currency detection YOLOv8 CV model trained on RBI-provided FICN datasets, deployable on mobile camera and bank counting machines
- Geospatial crime heatmap fraud complaint density mapping by district for law enforcement patrol prioritisation
- Fraud network graph transaction metadata and account linkage analysis for coordinated campaign detection
- Law enforcement command centre dashboard multi-agency intelligence sharing interface
- Court-admissible evidence packages full chain of custody with cryptographic audit trail

9. Setup and Deployment Guide

9.1 Local Development

Backend:

- cd backend
- python -m venv venv && venv\Scripts\activate
- pip install -r requirements.txt
- Create .env with GROQ_API_KEY, SUPABASE_URL, SUPABASE_KEY
- uvicorn main:app --reload

Frontend:

- cd frontend
- npm install
- npm run dev

9.2 Environment Variables Required

Variable	Description
GROQ_API_KEY	Groq API key for Llama 3.3 70B inference
SUPABASE_URL	Supabase project URL for community reports
SUPABASE_KEY	Supabase anon public key

9.3 Live Deployment

Component	Platform	URL
Frontend	Netlify	https://trinetra-india.netlify.app
Backend	Render	https://trinetra-backend-209a.onrender.com
API Docs	FastAPI Swagger	https://trinetra-backend-209a.onrender.com/docs
Source Code	GitHub	https://github.com/RithishPK/trinetra

10. Conclusion

Trinetra demonstrates that meaningful AI-powered public safety protection is achievable, deployable, and accessible to every Indian citizen today not in some future research paper.

The platform addresses the PS's core challenge: shifting from reactive case investigation to proactive threat detection. Every Indian with a smartphone can now paste a suspicious message, upload a suspicious document, or check a suspicious call script and receive an AI verdict with actionable guidance in under 3 seconds.

The combination of calibrated LLM prompting, deterministic verdict logic, multilingual support, community intelligence, and a court-ready audit trail positions Trinetra as a production-ready foundation for a national digital public safety infrastructure.

India lost Rs 1,776 crore to digital arrest scams in 2024. Trinetra is designed to make that number smaller.